

CSCI11 Final Exam REVIEW — Computer Architecture and Organization REVIEW

Multiple Choice — Write letter answer to the left of the question number (2 points each)

___ 1. Who is credited with developing the mathematical model of a device that can perform any computation?

- a. Charles Babbage
- b. Ada Lovelace
- c. Alan Turing
- d. John von Neumann

___ 2. According to Moore's Law, transistor count approximately doubles every:

- a. Six months
- b. One year
- c. Two years
- d. Five years

___ 3. What is the range of values a 3-bit unsigned integer can represent?

- a. 0 to 8
- b. -4 to 3
- c. 1 to 8
- d. 0 to 7

___ 4. What is the value of 2^{10} ?

- a. 512
- b. 256
- c. 2048
- d. 1024

___ 5. In two's complement, how do you obtain the negative of a positive number?

- a. Change only the most significant bit
- b. Flip every bit and add one
- c. Subtract from the maximum value
- d. Shift all bits one position to the right

___ 6. What is the two's complement representation of -3 in 4 bits?

- a. 1011
- b. 1110
- c. 1100
- d. 1101

___ 7. When does overflow occur in two's complement addition?

- a. When the carry-out bit is 1
- b. When the result is zero
- c. When both operands have the **same sign** AND the result has the **opposite sign**
- d. When either operand is negative

___ 8. The AND operation is most useful for which purpose?

- a. Setting bits
- b. Clearing bits
- c. Flipping bits
- d. Shifting bits

___ 9. What is the result of XOR performed on two identical bits?

- a. 1
- b. The original bit value
- c. 0
- d. Undefined

___ 10. What is the hexadecimal representation of binary 1100?

- a. B
- b. D
- c. C
- d. E

___ 11. De Morgan's Theorem states that $\text{NOT}(A \text{ OR } B)$ equals:

- a. $(\text{NOT } A) \text{ OR } (\text{NOT } B)$
- b. $\text{NOT}(A \text{ AND } B)$
- c. $A \text{ AND } B$
- d. $(\text{NOT } A) \text{ AND } (\text{NOT } B)$

- ___ 12. All computers can compute the same set of problems:
- If they have the same operating system
 - If they run the same programs
 - Given enough time and memory
 - They cannot — different architectures differ in capability
- ___ 13. A 2-input multiplexer is best described as a data:
- Distributor
 - Selector
 - Encoder
 - Decoder
- ___ 14. A n-input decoder converts an n-bit input to an output of how many bits?
- n bits
 - 2n bits
 - n^2 bits
 - 2^n bits
- ___ 15. What physical problems arise as transistor density increases?
- Signal interference and data loss
 - Memory corruption and clock drift
 - Voltage fluctuation and magnetic interference
 - Power consumption and heat generation
- ___ 16. After `JSR MYSUB` executes, what does R7 contain?
- The address of MYSUB
 - The address of the instruction before JSR
 - The value zero
 - The address of the instruction immediately after JSR
- ___ 17. Which LC-3 instruction uses base-plus-offset addressing?
- LD
 - LDR
 - LDI
 - LEA
- ___ 18. If data declarations occur before instructions at `.ORIG x3000`, what must the first executable statement be?
- `HALT`
 - Nothing — the assembler reorders code automatically
 - An unconditional `BR` targeting the first real instruction (e.g., `BR START`)
 - `NOP`
- ___ 19. PSR bit 15 = 1 indicates the CPU is running in:
- Interrupt mode
 - User mode
 - Supervisor mode
 - Halt mode
- ___ 20. A null-terminated string in LC-3 ends with the value:
- xFFFF
 - x0000
 - x00FF
 - x0020 (space)
- ___ 21. Which statement about `RTI` is correct?
- It is equivalent to `JMP R7`
 - It restores only the program counter
 - It can be safely executed in user mode
 - It pops both PC and PSR from the supervisor stack, restoring privilege and condition codes
- ___ 22. A subroutine called via `JSR` needs to call another subroutine with `JSR`. Before the inner `JSR`, the outer subroutine must:
- Clear R7 to zero
 - Do nothing — the hardware saves R7 automatically
 - Save R7 (the caller's return address) on the stack or in memory
 - Set R7 to the target address of the inner subroutine

___ 23. The trap vector table in LC-3 occupies addresses:

- a. x3000–x3FFF
- b. xFE00–xFFFF
- c. x0100–x01FF
- d. x0000–x00FF

___ 24. **STI R0, PTR** — when memory at PTR contains x5000 — stores R0 at:

- a. The address of PTR
- b. The address stored at x5000
- c. Address x0000
- d. Address x5000

___ 25. The **LD** instruction uses which addressing mode?

- a. PC-relative (PC + sign-extended offset)
- b. Register-indirect
- c. Base+offset (register + 6-bit offset)
- d. Immediate (literal encoded in the instruction)

___ 26. In the LC-3 calling convention, register R5 serves as:

- a. The stack pointer, tracking the top of the stack
- b. The return address register
- c. The frame pointer, providing stable offsets to local variables
- d. The global data pointer

___ 27. After a function completes its prologue in the LC-3 calling convention, the return value slot is located at:

- a. R5 + 0
- b. R5 + 1
- c. R5 + 3
- d. R5 + 2

___ 28. In the LC-3 calling convention, arguments are pushed onto the stack in which order?

- a. Left-to-right (first argument pushed first)
- b. Right-to-left (last argument pushed first)
- c. Alphabetical order by parameter name
- d. In order of data type size, smallest first

___ 29. In C, which keyword indicates that a function returns no value?

- a. null
- b. void
- c. none
- d. empty

___ 30. In C, what does passing a parameter "by value" mean?

- a. The function receives the address of the variable
- b. The function receives a pointer to a copy
- c. The function receives the actual variable and can change it in the caller
- d. The function receives a copy and changes do not affect the caller

___ 31. In the LC-3 calling convention, for a two-parameter function **add(a, b)**, parameter **a** (the first/left parameter) is located at:

- a. R5 + 3
- b. R5 + 2
- c. R5 + 5
- d. R5 + 4

___ 32. SIMD stands for:

- a. Sequential Instructions Multiple Data
- b. Simultaneous Instruction Memory Division
- c. Single Iteration Multiple Dispatch
- d. Single Instruction Multiple Data

___ 33. In C, the **%c** format specifier in **printf** is used to print:

- a. A decimal integer
- b. An ASCII character
- c. A floating-point number
- d. A hexadecimal value

___ 34. When a user-mode program makes a system call, the CPU must:

- a. Restart the program from the beginning
- b. Save all registers to external memory
- c. Transfer control directly to address x00FF
- d. Perform a mode switch from user mode to kernel mode

___ 35. In x86 assembly, `xor eax, eax` is preferred over `mov eax, 0` because:

- a. XOR executes faster on every CPU architecture
- b. It is a shorter encoding (2 bytes vs. 5 bytes) and CPUs recognize it as a zeroing idiom
- c. `MOV` cannot load the value zero
- d. XOR has higher precision for large integer values

Debug LC-3 Assembly — Identify the bug and explain the fix (5 points each)

1. The program is supposed to divide *first* by *second*. Which it appears to do (*times is correct*), however, the *remainder* in R1 is always negative. **What is the problem?**

```
                .orig x3000
                br start
first           .fill #47      ; first / second
second         .fill #5
start          LD  R2, second ;
                LD  R1, first  ; remainder in R1
                AND R0, R0, #0 ; number of times in R0
                NOT R2, R2
                ADD R2, R2, #1
subtr          ADD R1, R1, R2
                BRn end
                ADD R0, R0, #1
                BR subtr
end            HALT
                .end
```

2. The subroutine **GREET** is supposed to print **"Hi!"**, output a newline, and return to the caller. Instead the program hangs in an infinite loop after **GREET** is called. **What is the problem?**

; GREET: prints "Hi!" followed by a newline, then returns

```
GREET      LEA  R0, HI_MSG
            PUTS
            JSR  NEWLINE      ; print newline
            RET               ; return to original caller

NEWLINE    LD   R0, NL_CHAR
            OUT
            RET

HI_MSG     .STRINGZ "Hi!"
NL_CHAR    .FILL x000A      ; ASCII line feed
```

3. The following custom trap is supposed to clear R0 to zero and return control to the user program. Instead, execution terminates completely when the trap is called. **What is the problem?**

```
.ORIG x0026
.FILL CLEAR_TRAP
.END

.ORIG x0300
CLEAR_TRAP AND  R0, R0, #0      ; R0 = 0
            HALT               ; return to user program?
.END
```

4. This loop is supposed to sum all 6 elements of the array. Instead it sums only the first element and exits immediately. **What is the bug?**

```

        AND  R0, R0, #0      ; R0 = sum
        LD   R1, COUNT       ; R1 = 8
        LEA  R2, DATA
LOOP    LDR  R3, R2, #0      ; load element
        ADD  R0, R0, R3      ; accumulate
        ADD  R2, R2, #1      ; advance pointer
        ADD  R1, R1, #-1     ; decrement count
        BRn  LOOP           ; continue while count < 0???
        ST   R0, SUM
        HALT
COUNT  .FILL #6
DATA    .FILL #10
        .FILL #20
        .FILL #30
        .FILL #40
        .FILL #50
        .FILL #60
SUM     .BLKW 1
        .END

```

Binary Math — Write the answer to the right of each problem (2 points each)

All numbers are in **8-bit two's complement**. If the result overflows, write **overflow**.

1.

```

00110011
+ 00011001

```

2.

```

10100011
+ 11011100

```

3.

```

11010101
- 01001010

```

4.

```

01001010
+ 01000110

```

5.

```

10111100
+ 01011010

```

Program LC-3 Assembly (15 points each)

Points are awarded as follows per problem:

- **0–5 points** — Header comments: restate the problem, name registers and their roles
 - **0–5 points** — Correct syntax, neatness, and well-organized code
 - **0–5 points** — Algorithm is correct and the problem is solved
-

Problem 1

Write an LC-3 Assembly Language program that:

1. Uses a loop to traverse the string `TESTSTR` until the null terminator (`x0000`)
2. Counts the number of **uppercase letters** (ASCII `x41` through `x5A`, i.e., `A – Z`)
3. Stores the final count in the memory location `COUNT`, then `HALT`s

The following data declarations are provided — do not change them:

```
TESTSTR    .STRINGZ "Hello World"
COUNT     .BLKW 1
```

Hint: To test whether a character is in the range `x41–x5A`, subtract `x41`, check for negative (too low), then subtract 26 (`#26`), check for positive (too high). If neither, it is uppercase.

Provide answer on ruled paper.

Problem 2

Write a complete user-defined trap at vector `x0027` that finds the **largest value** in an array and returns the value in `R0`. Pass the address of the array in `R0` and the length of the array in `R1`. Assume the array contains signed integers. Save all registers used in trap. Use the array `VALUES` as your test case.

Provide **all three required sections**:

1. **Trap vector table entry**
2. **Handler code**
3. **Test program**

```
VALUES      .FILL #-3
             .FILL #7
             .FILL #0
             .FILL #5
             .FILL #-1
             .FILL #4
```

Provide answer on ruled paper.

Boolean Algebra — Write letter answer to the left of the question number (2 points each)

- ___ 1. What is the result of $A \text{ OR } A$?
- a. 0
 - b. NOT A
 - c. 1
 - d. A
- ___ 2. What is the result of $B \text{ AND } 1$?
- a. 0
 - b. NOT B
 - c. B
 - d. 1
- ___ 3. What is the result of $\text{NOT}(A \text{ OR } B)$?
- a. $(\text{NOT } A) \text{ AND } (\text{NOT } B)$
 - b. $A \text{ AND } B$
 - c. $(\text{NOT } A) \text{ OR } (\text{NOT } B)$
 - d. $A \text{ OR } B$
- ___ 4. What is the result of $A \text{ OR } (A \text{ AND } B)$?
- a. $A \text{ AND } B$
 - b. B
 - c. $A \text{ OR } B$
 - d. A
- ___ 5. What is the result of $(A \text{ AND } B) \text{ AND } 0$?
- a. A
 - b. 0
 - c. B
 - d. 1
- ___ 6. What is the result of $A \text{ AND } 0$?
- a. A
 - b. 1
 - c. NOT A
 - d. 0
- ___ 7. What is the result of $A \text{ XOR } 0$?
- a. 0
 - b. NOT A
 - c. A
 - d. 1
- ___ 8. What is the result of $\text{NOT}(\text{NOT } A)$?
- a. 0
 - b. 1
 - c. NOT A
 - d. A
- ___ 9. What is the result of $A \text{ AND NOT } A$?
- a. A
 - b. 1
 - c. 0
 - d. NOT A
- ___ 10. What is the result of $A \text{ OR NOT } A$?
- a. A
 - b. 0
 - c. NOT A
 - d. 1

Markdown (10 points total — 6 for accuracy, 4 for neatness)

In the space below, write the Markdown text to produce the following document. The parenthetical, italicized text describes the syntax required:

Course Notes (*heading 1*)

Introduction (*heading 2*)

References (*heading 2*) (*numbered list*)

Patt and Patel (*hyperlink* — use ***textbook.htm*** as the URL)

LC-3 Simulator (*hyperlink, file link* — use ***lc3tools.pdf*** as the filename)

Diagrams (*heading 2*) (*bulleted list*)

Circuit 1 (*image* — use ***circuit1.png*** as the image file)

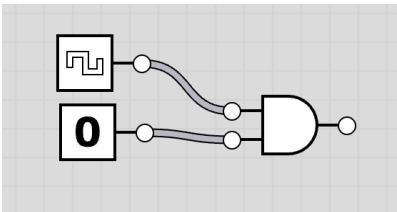
Circuit 2 (*image* — use ***circuit2.png*** as the image file)

Digital Design (4 points each)

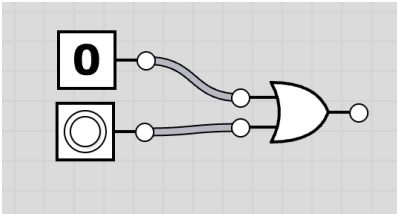
For each question:

1. Write your answer to the right of the image.
2. Provide the answer **AND** explain why.

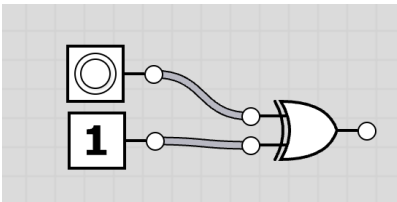
1. What is the output?



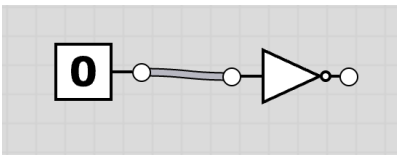
2. What is the output when the button is pressed?



3. What is the output when the button is pressed?



4. What is the output?



5. If $D = 1$, what is Q ?

